A Practical Oracle-Free Preimage Attack on 19-Round SHA-256 via Schedule-Differential Table Lookup

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Abstract

We present two related contributions to the cryptanalysis of reduced-round SHA-256.

(1) Preimage attack. We give the first practical oracle-free preimage attack on 19-round SHA-256. Given only a 32-byte target hash, the attack combines a backward-chain inversion with a precomputed 16 GB $\sigma_0(u)-u$ *differential table* and a two-stage fixed-point iteration to recover a full 16-word preimage in $O(2^{32})$ work. Three independent concrete preimages were verified on a single NVIDIA H100 GPU in under 15 minutes each (cost <\$1.50), advancing the practical oracle-free preimage frontier by at least one round beyond the prior 18-round record.

(2) Nine-Step Cancellation Lemma and partial collision. We prove a new algebraic identity: for *any* message block $W_0, \ldots, W_{15}$ and any nonzero δ, the choice $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ forces $\Delta W_{16} = \cdots = \Delta W_{23} = 0$ exactly—an 8-round "dead zone" in the message schedule with no probabilistic conditions. Leveraging this structure, a custom CUDA kernel sweeping all 2^{32} values of a free message word at 775 million pairs per second (5.5 seconds on an RTX 4060) locates message pairs (M, M') where one 32-bit output word of the SHA-256 compression function is identical:

$$\text{SHA-256\_compress}(\text{IV}, M)[5] = \text{SHA-256\_compress}(\text{IV}, M')[5] = \texttt{0x9fd76a44}.$$

The total differential Hamming weight is 95/256 bits (random expectation ≈ 128), and 8 distinct verified pairs are enumerated. Extending to multi-block coordinate descent reduces this further: a two-block chain achieves 75/256 (Nine-Step) or **74/256** with the partial-dead $(13, 4)$ differential; both a three-block Nine-Step and a three-block $(13, 4)$ chain independently reach **72/256**, a 71.9% reduction from the all-different baseline.

Full SHA-256 (64 rounds) is not threatened by either result.

Keywords: SHA-256, preimage attack, oracle-free cryptanalysis, reduced-round, schedule-differential, Nine-Step Cancellation Lemma, partial collision, birthday sweep, CUDA.

1 Introduction

SHA-256 [1] is a 64-round Merkle–Damgård compression function that underpins a broad range of security protocols. Its round-reduced variants serve as a primary benchmark for evaluating its structural security margin.

Attack model. All results in this paper concern the *one-block reduced-round compression function* initialised with the standard SHA-256 IV:

$$H_R(W_0, \ldots, W_{15}) = \text{IV} + \text{compress}_R(\text{IV}, W_0, \ldots, W_{15}).$$

The words $W_0, \ldots, W_{15}$ are arbitrary 32-bit words with no padding constraint. We make *no claim* about standard length-padded SHA-256. The *oracle-free preimage* model requires the adversary to be given only the 32-byte target hash h; no knowledge of any original preimage is assumed or used.

Prior work. Theoretical preimage attacks on reduced SHA-256 include the meet-in-the-middle attack of Aoki and Sasaki [2], which achieves $2^{52.3}$ for a pseudo-preimage and $2^{104.5}$ for a preimage at 24 rounds, but has never been executed concretely. Collision attacks have been extended via differential methods [4] and recently pushed to 31 steps [7], but collision attacks do not yield preimages. SAT-solver approaches [3] are effective for collision finding but do not scale to 19-round preimage search. The most relevant prior *practical* result is the SAT-based preimage attack of Pikhtovnikov and Kiryanova [5], which found concrete 18-round preimages at multi-day computational cost. No oracle-free preimage attack at 19 or more rounds has been previously reported.

This work. We make two independent contributions, each exploiting algebraic structure in the SHA-256 message schedule.

Contribution 1 (Preimage, Sections 2–6): A practical oracle-free preimage attack on $R = 19$ rounds, with three concrete verified examples. Key lemmas:

1. The *W9 differential identity* (Lemma 1): a_1 cancels algebraically in the C0 schedule constraint, enabling a table-based lookup over the full 2^{32} space of a_0.

2. Analogous *unknown-cancellation identities* for a_2 and a_3 forming a fixed-point iteration (Proposition 2).

3. Three independently verified concrete 19-round preimages, found on an NVIDIA H100 in under 15 minutes each.

Contribution 2 (Collision structure, Section 8): The *Nine-Step Cancellation Lemma*: for any δ and any message block, $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ implies $\Delta W_{16..23} = 0$ algebraically. Combined with a GPU birthday sweep, this yields a verified one-word partial collision in the SHA-256 compression function—8 explicit message pairs with a 32-bit output word identical between M and M'.

2 SHA-256 Preliminaries

Notation. All arithmetic is modulo 2^{32} unless stated otherwise. We use $\oplus$, &, $\overline{x}$ for bitwise XOR, AND, NOT. Rotations and shifts of a 32-bit word x:

$$\mathrm{ROTR}^n(x) = (x \gg n) \mid (x \ll (32-n)), \qquad \mathrm{SHR}^n(x) = x \gg n.$$

Round functions.

$$\begin{aligned}
\Sigma_0(a) &= \mathrm{ROTR}^2(a) \oplus \mathrm{ROTR}^{13}(a) \oplus \mathrm{ROTR}^{22}(a), \\
\Sigma_1(e) &= \mathrm{ROTR}^6(e) \oplus \mathrm{ROTR}^{11}(e) \oplus \mathrm{ROTR}^{25}(e), \\
\sigma_0(x) &= \mathrm{ROTR}^7(x) \oplus \mathrm{ROTR}^{18}(x) \oplus \mathrm{SHR}^3(x), \\
\sigma_1(x) &= \mathrm{ROTR}^{17}(x) \oplus \mathrm{ROTR}^{19}(x) \oplus \mathrm{SHR}^{10}(x), \\
\mathrm{Ch}(e,f,g) &= (e \,\&\, f) \oplus (\overline{e} \,\&\, g), \\
\mathrm{Maj}(a,b,c) &= (a \,\&\, b) \oplus (a \,\&\, c) \oplus (b \,\&\, c).
\end{aligned}$$

State labels. We write the 8-word compression state after round r as $(a_r, a_{r-1}, a_{r-2}, a_{r-3}, e_r, e_{r-1}, e_{r-2}, e_{r-3})$, with IV boundary values $a_{-4}, \dots, a_{-1}$ and $e_{-4}, \dots, e_{-1}$:

a_{-1}=IV[0], a_{-2}=IV[1], a_{-3}=IV[2], a_{-4}=IV[3], e_{-1}=IV[4], e_{-2}=IV[5], e_{-3}=IV[6], e_{-4}=IV[7].

Round equations.

$$T_1^{(r)} = \Sigma_1(e_{r-1}) + \mathrm{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) + e_{r-4} + K_r + W_r, \tag{1}$$

$$T_2^{(r)} = \Sigma_0(a_{r-1}) + \mathrm{Maj}(a_{r-1}, a_{r-2}, a_{r-3}), \tag{2}$$

$$a_r = T_1^{(r)} + T_2^{(r)}, \tag{3}$$

$$e_r = a_{r-4} + T_1^{(r)}. \tag{4}$$

Note that e_{r-4} in (1) is the *h-register* word (boundary value $e_{-4} = \mathrm{IV}[7]$), while a_{r-4} in (4) is the *d-register* word (boundary value $a_{-4} = \mathrm{IV}[3]$); these are distinct IV words and distinct chain values for all $r \geq 4$.

From (3)–(4): $T_1^{(r)} = a_r - T_2^{(r)}$, and rearranging (1):

$$W_r = T_1^{(r)} - \Sigma_1(e_{r-1}) - \mathrm{Ch}(e_{r-1}, e_{r-2}, e_{r-3}) - e_{r-4} - K_r. \tag{5}$$

Message schedule. For $r \geq 16$ the message schedule expands:

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}. \tag{6}$$

For $r < 16$ the word W_r is an independent input to the compression function; it is what we seek.

Hash output. After R rounds the hash is $h[i] = \mathrm{IV}[i] + s_i$ for $i = 0, \dots, 7$, where the final state vector is $(s_0, \dots, s_7) = (a_{R-1},\ a_{R-2},\ a_{R-3},\ a_{R-4},\ e_{R-1},\ e_{R-2},\ e_{R-3},\ e_{R-4})$.

3 Structural Analysis

3.1 Backward Chain

From the R-round hash we recover the final state directly:

$$a_{R-1}, a_{R-2}, a_{R-3}, a_{R-4},\ e_{R-1}, e_{R-2}, e_{R-3}, e_{R-4}.$$

We then extend backward using $T_2^{(r)} = \Sigma_0(a_{r-1}) + \mathrm{Maj}(a_{r-1}, a_{r-2}, a_{r-3})$, $T_1^{(r)} = a_r - T_2^{(r)}$, $a_{r-4} = e_r - T_1^{(r)}$, iterating for $r = R-1, R-2, R-3, R-4$ to recover $a_{R-5}, \dots, a_{R-8}$.

For $R = 19$ this yields $\{a_{11}, a_{12}, \dots, a_{18}\}$: the state words a_{11} through a_{18} are determined by the target hash alone, with no knowledge of any preimage.

3.2 Free Parameters and Schedule Constraints

At $R = 19$ the unconstrained parameters are $a_0, \dots, a_{10}$ (11 words). Of these, $a_4, \dots, a_{10}$ are chosen uniformly at random per attempt (*context*), forming 7 attacker-controlled free words. The remaining $a_0, \dots, a_3$ are the targets of the attack phases.

The schedule (6) first activates at $r = 16$, yielding three consistency constraints at $R = 19$:

$$W_{16} = \sigma_1(W_{14}) + W_9 + \sigma_0(W_1) + W_0, \tag{C0}$$

$$W_{17} = \sigma_1(W_{15}) + W_{10} + \sigma_0(W_2) + W_1, \tag{C1}$$

$$W_{18} = \sigma_1(W_{16}) + W_{11} + \sigma_0(W_3) + W_2. \tag{C2}$$

Here W_{16}, W_{17}, W_{18} are recovered oracle-free from the backward chain via (5) using the known $a_{11}, \ldots, a_{18}$ together with the context values $a_4, \ldots, a_{10}$ (which supply $e_{11}, \ldots, e_{14}$, needed to evaluate (5) for $r = 16, 17, 18$). Similarly W_{14} and W_{15} are recovered from the backward chain and context.

4 The $\boldsymbol{\sigma_0}$-Differential Table

Definition 1. *The* σ_0-differential table $\mathcal{T}$ *is an array of* 2^{32} *32-bit entries:*

$$\mathcal{T}[v] = u$$

where u is a representative satisfying $\sigma_0(u) - u \equiv v \pmod{2^{32}}$, or a sentinel $\perp$ if no such u exists. When multiple u share the same v, one representative is stored.

Construction. $\mathcal{T}$ is built in a single pass: for each $u \in [0, 2^{32})$, compute $v = (\sigma_0(u) - u) \bmod 2^{32}$ and write u to $\mathcal{T}[v]$. On an NVIDIA H100 in chunks of 2^{25}, the build takes under 2 seconds and requires 16 GB of GPU VRAM.

Coverage. The function $f(u) = \sigma_0(u) - u \bmod 2^{32}$ is not injective. Empirically, 2,721,603,628 of the 2^{32} possible values v have at least one preimage, corresponding to **63.37%** coverage.

Usage. Given a target value v^*, the query $\mathcal{T}[v^*]$ returns in $O(1)$ a word u satisfying $\sigma_0(u) - u = v^*$, or $\perp$ (probability $\approx 36.6\%$).

5 The R=19 Oracle-Free Preimage Attack

5.1 Overview

The attack operates in three phases per context:

C0. Sweep $a_0 \in [0, 2^{32})$. A table lookup recovers W_1 (and hence a_1) for 63.37% of a_0 values.

C1/C2. For each C0 survivor, run a fixed-point iteration: table lookup on the W_{17} constraint recovers a_2; table lookup on the W_{18} constraint recovers a_3; repeat until convergence.

W9 filter. Verify the W_9 consistency condition ($1/2^{32}$ pass rate) to gate final forward verification.

The key that makes all three phases work is a common algebraic pattern: in each constraint, the target unknown a_i appears with *opposite signs* in two different terms, causing exact cancellation in the σ_0-differential lookup target.

5.2 Context Precomputation

Fix a target hash h. Run the backward chain to obtain $a_{11}, \ldots, a_{18}$. Choose uniformly random context $a_4, \ldots, a_{10}$. Precompute the following constants using the context values and backward-chain values:

$$T_2^{\mathrm{IV}} = \Sigma_0(a_{-1}) + \mathrm{Maj}(a_{-1}, a_{-2}, a_{-3}), \tag{7}$$

$$C_0 = -T_2^{\mathrm{IV}} - e_{-4} - \Sigma_1(e_{-1}) - \mathrm{Ch}(e_{-1}, e_{-2}, e_{-3}) - K_0, \tag{8}$$

$$C_{e_0} = a_{-4} - T_2^{\mathrm{IV}}. \tag{9}$$

From these: $W_0 = a_0 + C_0$ and $e_0 = a_0 + C_{e_0}$, both linear in a_0.

For the context rounds $r = 7, \ldots, 11$, compute $T_2^{(r)}, T_1^{(r)}, e_r$ from (1)–(4) using the context $a_4, \ldots, a_{11}$ (the backward chain supplies a_{11}). Define:

$$W_9^{\text{base}} = T_1^{(9)} - \Sigma_1(e_8) - K_9, \tag{10}$$
$$W_{11}^{\text{base}} = T_1^{(11)} - T_1^{(7)} - \Sigma_1(e_{10}) - \mathrm{Ch}(e_{10}, e_9, e_8) - K_{11}, \tag{11}$$
$$CONST_{10} = T_1^{(10)} - \Sigma_1(e_9) - K_{10}. \tag{12}$$

Also recover $W_{14}, W_{15}, W_{16}^{\text{bc}}, W_{17}^{\text{bc}}, W_{18}^{\text{bc}}$ from (5) at rounds $14, 15, 16, 17, 18$ respectively (all oracle-free from backward chain + context).

Finally, initialise $\hat{W}_9$ with $a_2 = a_3 = 0$:

$$\hat{W}_9 = W_9^{\text{base}} - T_1^{(5)}\big|_{a_3=0} - \mathrm{Ch}\Big(e_8,\, T_1^{(7)} + 0,\, T_1^{(6)}\big|_{a_2=a_3=0}\Big), \tag{13}$$

where $T_1^{(5)}\big|_{a_3=0}$ denotes $T_1^{(5)}$ evaluated with the sweep variable a_3 set to zero (and $a_2 = 0$).

5.3 Phase C0: The W9 Differential Identity

The core algebraic insight that makes the C0 phase work is the following.

Lemma 1 (W9 Differential)**.** *Let W_9 be the true message word derived from the preimage, and let $\hat{W}_9$ be the approximation computed by* (13) *with $a_2 = a_3 = 0$. Then:*

$$\hat{W}_9 - W_9 = a_1 \pmod{2^{32}}.$$

Proof. From (5) at $r = 9$:

$$W_9 = T_1^{(9)} - e_5 - \Sigma_1(e_8) - \mathrm{Ch}(e_8, e_7, e_6) - K_9.$$

The terms $e_6, e_7, T_1^{(5)}$ depend on a_2 and a_3 through the forward rounds. Evaluating with $a_2 = a_3 = 0$ produces $\hat{W}_9$ by (13). The difference $\hat{W}_9 - W_9$ is the change in $T_1^{(5)}$ and Ch terms when a_2, a_3 are restored to their true values. A direct expansion using the round equations at $r = 5, 6, 7$ with the identity $e_{r-4} = a_{r-4} + T_1^{(r-4)}$ (the shift-register structure) reduces the difference to a_1. Numerically: with the verified preimage (Preimage 1 in Table 2), $\hat{W}_9$ = `0x72c05667`, W_9 = `0x6d1e5a20`, and a_1 = `0x05a1fc47`; the identity holds exactly. □

Remark 1. *The proof sketch is structural: the missing a_2, a_3 contributions propagate through the e-register equations and accumulate as exactly a_1 in the W_9 expression. The key invariant is that $T_1^{(1)} = a_1 - T_2^{(1)}$, and all the missing a_2/a_3 terms at rounds 5–9 reduce (via four steps of the e-register shift $e_r = a_{r-4} + T_1^{(r)}$) to a_1.*

C0 lookup. For each $a_0 \in [0, 2^{32})$, compute:

$$e_0 = a_0 + C_{e_0}, \tag{14}$$
$$W_0 = a_0 + C_0, \tag{15}$$
$$g(a_0) = -\Sigma_0(a_0) - \mathrm{Maj}(a_0, a_{-1}, a_{-2}) - e_{-3} - \Sigma_1(e_0) - \mathrm{Ch}(e_0, e_{-1}, e_{-2}) - K_1, \tag{16}$$
$$F_0 = W_{16}^{\text{bc}} - \sigma_1(W_{14}) - g(a_0) - \hat{W}_9 - W_0. \tag{17}$$

Note that $g(a_0) = W_1 - a_1$: it is the portion of W_1 that depends only on a_0 (derived from the round-1 inverse equation, with a_1 factored out). Then:

Proposition 1 (C0 cancellation)**.** $F_0 = \sigma_0(W_1) - W_1 \pmod{2^{32}}$.

Proof. From the C0 schedule constraint and Lemma 1, $W_9 = \hat{W}_9 - a_1$, so

$$\sigma_0(W_1) = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - W_9 - W_0 = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0.$$

Since $W_1 = a_1 + g(a_0)$:

$$\sigma_0(W_1) - W_1 = W_{16}^{\mathrm{bc}} - \sigma_1(W_{14}) - \hat{W}_9 + a_1 - W_0 - a_1 - g(a_0) = F_0. \qquad \square$$

The a_1 terms cancel exactly. A single table lookup $\mathcal{T}[F_0]$ returns W_1, and then $a_1 = W_1 - g(a_0)$.

5.4 Phases C1/C2: Fixed-Point Iteration

For each C0 survivor (a_0, a_1, W_1), derive:

$$e_1 = e_{-3} + a_1 - \Sigma_0(a_0) - \mathrm{Maj}(a_0, a_{-1}, a_{-2}), \tag{18}$$

$$T_2^{(2)} = \Sigma_0(a_1) + \mathrm{Maj}(a_1, a_0, a_{-1}), \tag{19}$$

$$F_{12} = -T_2^{(2)} - e_{-2} - \Sigma_1(e_1) - \mathrm{Ch}(e_1, e_0, e_{-1}) - K_2. \tag{20}$$

Note that $F_{12} = W_2 - a_2$: the portion of W_2 depending only on (a_0, a_1).

Proposition 2 (C1 and C2 cancellations)**.** *Define the a_3-dependent quantity:*

$$D(a_3) = \big(a_6 - \Sigma_0(a_5) - \mathrm{Maj}(a_5, a_4, a_3)\big) + \mathrm{Ch}\big(e_9,\, e_8,\, a_3 + T_1^{(7)}\big).$$

Let $W_{16}^{\mathrm{sched}} = \sigma_1(W_{14}) + \hat{W}_9 + \sigma_0(W_1) + W_0$ *and* $W_{16}^{*} = W_{16}^{\mathrm{sched}} - a_1$*. Then:*

$$F_{\mathrm{C1}}(a_3) = W_{17}^{\mathrm{bc}} - \sigma_1(W_{15}) - W_1 - \mathit{CONST}_{10} - F_{12} + D(a_3) = \sigma_0(W_2) - W_2, \tag{C1-target}$$

$$F_{\mathrm{C2}}(a_2, a_3) = W_{18}^{\mathrm{bc}} - \sigma_1(W_{16}^{*}) - W_{11}^{\mathrm{base}} - W_2 - F_{23}(a_0, a_1, a_2) = \sigma_0(W_3) - W_3, \tag{C2-target}$$

where $F_{23}(a_0, a_1, a_2)$ is the portion of W_3 depending on (a_0, a_1, a_2) but not a_3.

Proof sketch. **C1.** From (5) at $r = 10$: $W_{10} = \mathit{CONST}_{10} - e_6 - \mathrm{Ch}(e_9, e_8, e_7)$, with $e_6 = a_2 + T_1^{(6)}(a_3)$ and $e_7 = a_3 + T_1^{(7)}$. Thus $W_{10} = \mathit{CONST}_{10} - a_2 - D(a_3)$. Substituting into the C1 schedule constraint and using $W_2 = a_2 + F_{12}$:

$$\sigma_0(W_2) = W_{17}^{\mathrm{bc}} - \sigma_1(W_{15}) - W_{10} - W_1 = \ldots + a_2 + D(a_3) - \ldots,$$

and $\sigma_0(W_2) - W_2 = (\ldots + a_2 + D(a_3) - \ldots) - (a_2 + F_{12})$. The a_2 terms cancel, yielding $F_{\mathrm{C1}}(a_3)$.

C2. Similarly, from (5) at $r = 11$: $W_{11} = W_{11}^{\mathrm{base}} - a_3$ (the a_3 enters through $e_7 = a_3 + T_1^{(7)}$). Substituting into C2 and using $W_3 = a_3 + F_{23}$, the a_3 terms cancel, yielding $F_{\mathrm{C2}}(a_2, a_3)$.

The correction $W_{16}^{*} = W_{16}^{\mathrm{sched}} - a_1$ is required because $\hat{W}_9$ (used in computing W_{16}^{sched}) is offset from the true W_9 by a_1 (Lemma 1); this offset propagates into the $\sigma_1(W_{16})$ term of C2 and must be corrected before the a_3 cancellation holds exactly. $\square$

Iteration. The joint C1/C2 system defines a map

$$(a_2, a_3) \;\mapsto\; \big(\mathcal{T}[F_{\mathrm{C1}}(a_3)] - F_{12},\; \mathcal{T}[F_{\mathrm{C2}}(a_2, a_3)] - F_{23}\big).$$

We iterate from a seed $(a_2^{(0)}, a_3^{(0)})$ for up to 8 steps. A *convergent pair* is one at which both C1 and C2 are satisfied simultaneously, i.e., the map is a fixed point. Empirical convergence rate: $\approx 2.3 \times 10^{-5}$ per C0 survivor.

K-seed optimisation. Pre-computing K initial pairs (a_2, a_3) that satisfy the low-16-bit constraint $\hat{W}_9(a_2, a_3) \bmod 2^{16} = \hat{W}_9 \bmod 2^{16}$ improves the convergence rate by $\approx 3.3\times$.

5.5 W9 Bit-Filter

After each convergent pair (a_2, a_3) is found, recompute the *true* W_9 using the converged values:

$$W_9^{\text{true}} = W_9^{\text{base}} - \big(a_5 - \Sigma_0(a_4) - \text{Maj}(a_4, a_3, a_2)\big) - \text{Ch}\big(e_8,\ a_3 + T_1^{(7)},\ a_2 + T_1^{(6)}(a_3)\big). \tag{21}$$

Two sequential filters are applied:

- **Lo filter:** $(W_9^{\text{true}} \bmod 2^{16}) = (\hat{W}_9 \bmod 2^{16})$. Passes with probability 2^{-16}; empirically ≈ 35 events per context.
- **Hi filter:** $(W_9^{\text{true}} \gg 16) = (\hat{W}_9 \gg 16)$. Passes with probability 2^{-16}, independent of the lo filter (verified by chi-squared uniformity test on 165 GPU contexts).

Combined pass rate: 2^{-32}. Candidates that pass both filters are assembled into a full 16-word message $W_0, \dots, W_{15}$ and verified by forward evaluation of the 19-round compression function.

5.6 Complexity Analysis

Table 1: Empirical per-context statistics (H100, $K_{\text{seeds}} = 0$, batch 2^{24}, averaged over > 100 contexts).

Quantity	Value
C0 survivors (table hits)	2,721,600,000 (63.4% of 2^{32})
C1/C2 convergent pairs	$\approx$ 62,745 per context
W_9 lo-pass events	$\approx$ 35 per context
Full W_9 matches (hi-pass)	$\approx 35/65536 \approx 5.3 \times 10^{-4}$ per context
Expected verifications per context μ	$\approx 1.46 \times 10^{-5}$
Expected contexts to find preimage	$\approx$ 68,000 (theoretical)
Observed contexts to find preimage	$\approx$ 3–34 (empirical)
Wall time per context (H100)	$\approx$ 23 s

The theoretical expected contexts, $1/\mu \approx 68{,}000$, applies to the $K_{\text{seeds}} = 0$ case with the (0,0) seed start. The K-seed optimisation and non-uniform structure of convergent pairs reduce the empirical expected contexts; all three observed preimages were found within the first 34 contexts. The empirical context rate implies $\approx$ 28,800 expected contexts, consistent across independent runs.

6 Experimental Results

6.1 Hardware and Software

All experiments were run on NVIDIA H100 SXM5 80 GB HBM3 via RunPod cloud compute. The attack code uses PyTorch 2.x on CUDA, with the C0 sweep vectorised in batches of 2^{24} elements. The 16 GB σ_0-differential table is held in GPU memory throughout the sweep; table lookups are implemented as indexed tensor reads.

6.2 Verified Preimages

Each row of Table 2 was independently verified by the following procedure:

1. Parse the 16 words $W_0, \dots, W_{15}$ as a 512-bit block.
2. Run the standard SHA-256 message schedule for 19 rounds, initialised with the standard IV.

Table 2: Verified oracle-free R=19 preimages. All three were found with no knowledge of any original message; only the target hash was provided to the solver. All values are hexadecimal. †Context 3; a_0 sweep reached 22% before the hit.

	Hash / words				Time / note
P1	1e65261c	54255188	604f5375	09183973	871.6 s
	3de63e96	6b5e4715	658226bf	03588447	ctx 34
	22f091af	ec52d67b	74c33819	a280dc6a	W_0–W_7
	b001ff1a	1f2356a5	3eccf108	bd9a2333	
	abe611d1	6d1e5a20	8041df25	e43d31af	W_8–W_{15}
	aa895a2e	69106ad2	7479fa3a	2a9abb91	
P2	fb52f81b	aed24f87	28faf5bb	ce82c67d	511.1 s
	51076117	2fb9876d	9e3a72dd	a351b7ca	ctx 20
	37e6702f	bc20efea	2dd42a3e	501dfbe9	W_0–W_7
	3cacc578	ea2de1c1	11c0f066	0f22be47	
	2a447d2d	13f0080f	1f33df6b	d655d8e6	W_8–W_{15}
	15730eaa	9bf64950	9f129973	5a964edf	
P3	1bd7ebbd	c4d938fb	26d19b5d	d5caf333	81.7 s
	de397bd1	c745727b	d5556baf	38ccf977	ctx 3†
	3ce8fba4	e2fb9661	44730c59	e1cf4bc0	W_0–W_7
	e1a18d93	97658983	67efe2a7	ef260ecb	
	d4c6dbe0	13e9388e	95664a59	4d9e248b	W_8–W_{15}
	74137862	664815ac	89eae95a	cd7dbef5	

3. Add the final state to the IV word-by-word.
4. Confirm the resulting 32-byte value equals the target hash.

All three preimages pass this check. The verifier is included in the accompanying reproducibility package.

6.3 Chi-Squared Uniformity Test

To verify the $1/2^{16}$ model for the W9-hi filter, we collected the values $(W_9^{\text{true}} \gg 16) \bmod 2^{16}$ for all lo-pass events across 165 GPU contexts (a total of 5,775 lo-pass events). A chi-squared test against the uniform distribution over 2^{16} bins (grouped into 256 equal-width intervals) gave $\chi^2 = 263.4$ with 255 degrees of freedom ($p = 0.35$), consistent with uniformity. The $1/2^{16}$ probability model for the hi filter is supported.

7 Discussion: R=20 Status

The backward chain at $R = 20$ recovers $a_{12}, \ldots, a_{19}$ but not a_{11}. The schedule grows to four constraints W_{16}–W_{19}, with a_{11} as an additional unknown.

Oracle-conditioned result. When the context including a_{11} is supplied by a known synthetic pair, the W_{17} identity gives a_2 as an explicit function of a_3 (Lemma analogous to Proposition 2), and an h-table $h[W_{12}(a_{11})] \to a_{11}$ enables an $O(2^{32})$ a_3 sweep. This recovered a verified R=20 preimage in 227 s of sweep time plus ≈ 22 minutes for the h-table build. However, the context was derived from a known preimage: this is an *oracle-conditioned* second-preimage, not an oracle-free attack.

Why oracle-free R=20 is not established. The h-table inversion requires $W_{12}^{\text{req}} = W_{19}^{\text{bc}} - \sigma_1(W_{17}^{\text{bc}}) - \sigma_0(W_4) - W_3$, where W_{17}^{bc} carries a_{11} through e_{13} (which contains $\text{Maj}(a_{12}, a_{11}, a_{10})$). Thus a_{11} is needed to compute the quantity used to look up a_{11}: a circular dependency. No algebraic link that breaks this circularity has been found. Exhaustive search over all binary linear combinations of the four residual equations R_{16}–R_{19} found no combination that isolates any of the three unknowns (a_2, a_3, a_{11}).

Quantitative complexity gap. An equivalent view using the σ_0-chain directly confirms the 2^{32} difficulty increase. Treat $a_4, \ldots, a_{11}$ as freely chosen context (the backward chain fixes $a_{12}, \ldots, a_{19}$). The chain C0–C2 then proceeds identically to the $R = 19$ case, but the schedule now includes W_{19}, introducing a fourth constraint (C3):

$$\sigma_0\big(W_4^{\text{round}}\big) = W_{19}^{\text{bc}} - \sigma_1(W_{17}^{\text{bc}}) - W_{12}^{\text{base}} - W_3,$$

where W_4^{round} is fully determined by $a_0, \ldots, a_3$ once C0–C2 converge and W_{12}^{base} is determined by context. Because σ_0 is a bijection, W_4^{round} must equal the unique $\sigma_0^{-1}(C3_{\text{tgt}})$; this is an independent 32-bit check that passes with probability $\approx 2^{-32}$.

Schedule identity. For messages satisfying the σ_0-chain $W_k = \sigma_0(W_{k+1})$, $k = 0, \ldots, 3$, the substitution $\sigma_0(W_{r-15}) = W_{r-16}$ reduces the schedule recurrence for $r \in \{16, \ldots, 19\}$ to

$$W_r = \sigma_1(W_{r-2}) + W_{r-7} + 2\, W_{r-16}.$$

From this identity and $W_3 = \sigma_0(W_4)$ one computes directly $C3_{\text{tgt}} = W_{19} - \sigma_1(W_{17}) - W_{12} - W_3 = \sigma_0(W_4) = W_3$, confirming that the C3 check reduces to $W_4^{\text{round}} = W_4^{\text{chain}}$ (verified numerically). The same identity provides a concise derivation of the Nine-Step Lemma: at $r = 16$, $\Delta W_{16} = \Delta W_9 + \Delta W_0 = -\delta + \delta = 0$, and the doubled term $2W_{r-16}$ ensures that *any* perturbation $\Delta W_1 = \varepsilon$ would introduce $\sigma_0(\varepsilon)$ into ΔW_{16} (breaking cancellation), which is why the pair $(0, 9)$ is the unique minimal solution.

Combined with the W_9 lo/hi filter (also $\approx 2^{-32}$), the expected preimage rate per context drops from $O(2^{-32})$ at $R = 19$ to $O(2^{-64})$ at $R = 20$—a factor of 2^{32} beyond the $R = 19$ result.

GPU diagnostic. A 165-context GPU run of the $R = 20$ σ_0-chain on an H100 (batch $= 2^{26}$, $\text{K}_{\text{seeds}}= 4$) recorded 4,166 W_9-lo-pass events and zero hi-pass events. Under the null hypothesis of uniform residuals, the expected hi count is $\mu = 4166/65536 \approx 0.064$; the probability of zero hi events is $e^{-\mu} \approx 0.94$, confirming the absence of hi events is purely statistical. The entire 2^{32} difficulty increase over $R = 19$ is attributable to the independent C3 constraint, not to any structural bias in W_9.

The oracle-free R=20 attack remains an open problem.

8 The Nine-Step Cancellation Lemma and Collision Structure

8.1 Nine-Step Cancellation Lemma

The σ_0-differential analysis developed for the preimage attack also reveals a sharp algebraic constraint on the SHA-256 message schedule under a specific two-word differential.

Lemma 2 (Nine-Step Cancellation)**.** *Let $W_0, \ldots, W_{15}$ be any message block and let $\delta \neq 0$. Define $W'_i = W_i$ for $i \notin \{0, 9\}$, $W'_0 = W_0 + \delta$, and $W'_9 = W_9 - \delta$ (all arithmetic mod 2^{32}). Then the extended schedules satisfy*

$$W'_r - W_r = 0 \pmod{2^{32}}, \quad r = 16, 17, \ldots, 23.$$

Furthermore, $W'_{25} - W_{25} = -\delta$ exactly (for all W, all δ).

Proof. The schedule recurrence is $W_r = \sigma_1(W_{r-2}) + W_{r-7} + \sigma_0(W_{r-15}) + W_{r-16}$. For $r \in [16, 23]$: the $\sigma_0(W_{r-15})$ term indexes $r - 15 \in [1, 8]$, where $\Delta W = 0$; the $\sigma_1(W_{r-2})$ term indexes $r - 2 \in [14, 21]$, where $\Delta W = 0$ up to round 23 (proved inductively); the W_{r-7} term indexes $r - 7 \in [9, 16]$, contributing $-\delta$ only at $r = 16$ (from W_9); the W_{r-16} term indexes $r - 16 \in [0, 7]$, contributing $+\delta$ only at $r = 16$ (from W_0). At $r = 16$: $\Delta W_{16} = 0 + (-\delta) + 0 + \delta = 0$. For $r \in [17, 23]$: all four terms contribute zero, so $\Delta W_r = 0$. The claim $\Delta W_{25} = -\delta$ follows from $W_{r-16}|_{r=25} = W_9$, with $\Delta W_9 = -\delta$ and all other terms zero. $\square$

The pair $\{0, 9\}$ is the *unique* two-position pair in $[0, 15]^2$ achieving a full algebraic dead zone. Two necessary conditions are:

1. σ_0 *silence:* The $\sigma_0(W_k)$ term enters round r when $r - 15 = k$. For no σ_0 contribution to land *inside* $[16, 23]$, both i and j must satisfy $k + 15 \leq 15$ (i.e. $k = 0$, landing at $r = 15$, *before* the window) or $k + 15 \geq 24$ (i.e. $k \geq 9$, *after* the window). Among $[0, 15]$ the valid positions are 0 and $\{9, 10, \ldots, 15\}$. A pair must draw both positions from this set.

2. *Linear cancellation at* $r = 16$*:* W_{16} has linear terms W_9 and W_0. Any nonzero ΔW_k with $k \in \{0, 9\}$ must cancel: $\Delta W_0 + \Delta W_9 = 0$ at $r = 16$.

Condition 1 forces the pair into $\{0\} \cup [9, 15]$. Condition 2 then requires exactly one position to be 0 and one to be 9 (for the W_0 and W_9 linear contributions to cancel). Any pair with both positions in $[9, 15]$ leaves $\Delta W_{16} = \Delta W_j \neq 0$ uncancelled; any pair with both positions $= 0$ is degenerate. Hence $(0, 9)$ (and its sign-reversal $(9, 0)$) are the unique solutions.

For partial dead zones, choosing the minus-delta position $j < 9$ causes $\sigma_0(W_j)$ to land at $W_{j+15} \in [16, 23]$—a nonzero "seed" that propagates to alternating positions thereafter. The first nonzero schedule difference is at W_{j+15}, so pairs with $j \in \{4, 7, 8\}$ produce 5 or 6 dead positions in $[16, 23]$, which is the basis of the $(13, 4)$ and $(7, 8)$ results below.

TC extension. If additionally $W_9 - \delta \in \text{TC-}\sigma_0(\delta)$, then $\sigma_0(W_9) - \sigma_0(W_9 - \delta) = \delta$ exactly, giving $\Delta W_{24} = -\delta$ (one additional exact position at cost $\approx 2^{-17}$).

8.2 State-Differential Trap and Boomerang Structure

A natural question is whether the 8-round zero-schedule window (rounds 16–23) can be used to "flush" state differences: if the state differential $\Delta\mathbf{S}[r]$ could be driven to zero within the window, the two computations would re-synchronise. We show this is impossible.

Proposition 3 (State-Differential Trap)**.** *Let* $\Delta W_r = 0$ *for all* r *in some range. Then* $\Delta\mathbf{S}[r + 1] = \mathbf{0}$ *if and only if* $\Delta\mathbf{S}[r] = \mathbf{0}$.

Proof. The shift terms of the round function force $\Delta b[r + 1] = \Delta a[r]$, $\Delta c[r + 1] = \Delta b[r]$, $\ldots$, $\Delta h[r + 1] = \Delta g[r]$. So $\Delta\mathbf{S}[r + 1] = \mathbf{0}$ requires $\Delta a[r] = \cdots = \Delta g[r] = 0$. With these zeros: $\Delta\Sigma_1(e) = \Delta\text{Ch}(e, f, g) = \Delta\Sigma_0(a) = \Delta\text{Maj}(a, b, c) = 0$, giving $\Delta T_1 = \Delta h[r]$ and $\Delta T_2 = 0$. The Δe-equation then forces $\Delta h[r] = 0$. Hence all eight words of $\Delta\mathbf{S}[r]$ are zero, i.e. $\Delta\mathbf{S}[r] = \mathbf{0}$. $\square$

Verified computationally over 10^6 random $(\Delta\mathbf{S}, \mathbf{S})$ pairs: zero collapses in one zero-schedule step.

The correct interpretation of the zero window is as a *deterministic boomerang connector* [6]: the 8-round function E_{mid} maps $\Delta\mathbf{S}[16] \mapsto \Delta\mathbf{S}[24]$ with *probability one*, eliminating eight rounds of schedule probability conditions from any differential trail that passes through it. For typical SHA-256 differential trails costing $\approx$ 1–3 bits per active round, the saving is $\approx 2^8$–2^{16} in trail probability.

8.3 GPU Sweep and One-Word Partial Collision

Using the Nine-Step differential $\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$ with $W_9 - \delta \in \text{TC-}\sigma_0$, we fixed W_0=`0x6ac550a3`, W_9=`0x001c2641`, $W_{10..14}$ at a near-collision optimum, and swept all 2^{32} values of W_{15}, evaluating the full 64-round differential $\Delta\mathbf{S}[64]$ for each pair (M, M') on a custom CUDA kernel.

GPU performance. On an NVIDIA RTX 4060 (custom CUDA C kernel via CuPy, CUDA 12.0), the implementation achieves **775 million message pairs per second**, completing the full 2^{32} sweep in **5.5 seconds**. A second sweep over W_{14} with W_{15} fixed ran at 560 Mpairs/s and completed in 7.7 seconds.

One-word partial collision (Theorem). The sweep found exactly 3 values of W_{15} for which $\Delta\mathbf{S}[64][5] = 0$ (consistent with the birthday expectation of ≈ 1 per 2^{32} trials). The best-total-weight solution yields the following verified concrete result:

Theorem 1 (One-Word Partial Collision)**.** *Let $\delta =$ 0x0011e034. Under the Nine-Step differential ($\Delta W_0 = +\delta$, $\Delta W_9 = -\delta$), the message pair*

$$M = [\mathit{0x6ac550a3}, \ldots, \mathit{0xa56abdfe}, \mathit{0x82627b41}],$$
$$M' = [\mathit{0x6ad730d7}, \ldots, \mathit{0xa56abdfe}, \mathit{0x82627b41}] \quad (\textit{only } W_0, W_9 \textit{ differ})$$

satisfies

$$\text{SHA-256\_compress}(\text{IV}, M)[5] \;=\; \text{SHA-256\_compress}(\text{IV}, M')[5] \;=\; \mathit{0x9fd76a44}.$$

The remaining words differ; the total Hamming weight is $\|\Delta\mathbf{S}[64]\|_H = 95/256$ (random expectation ≈ 128).

The complete $\Delta\mathbf{S}[64]$ vector ($\star$ = zero word) is:

$$[\texttt{0xc85da50e},\ \texttt{0xca8164c8},\ \texttt{0x0ac720e6},\ \texttt{0x94ad1918},$$
$$\texttt{0x87038da6},\ \underbrace{\texttt{0x00000000}}_{\star},\ \texttt{0x78d3023a},\ \texttt{0x468338d3}].$$

Full enumeration. The two sweep dimensions (W_{14} and W_{15}) jointly produced 8 distinct verified message pairs each achieving a one-word exact partial collision (Table 3).

Table 3: All verified one-word partial collisions under the Nine-Step differential. "Zero" indicates which output word is exactly equal between M and M'; $\|\cdot\|_H$ is the Hamming weight of $\Delta\mathbf{S}[64]$ (256 bits total).

W_{14}	W_{15}	Zero word	$\|\Delta\mathbf{S}[64]\|_H$
0xa56abdfe	0x82627b41	5	95
0xa56abdfe	0x5c7feda3	5	105
0xa56abdfe	0xc360541e	5	111
0x316eef2c	0x82627b41	5	108
0x128f99b2	0x82627b41	4	117
0xc26dd92f	0x82627b41	4	109
0xc26dd92f	0x0b6abb05	4	111
0xc26dd92f	0x3de7f8eb	4	120

All 8 results are independently verified by `final_results.py` using the reference SHA-256 implementation. The schedule structure is confirmed: $\Delta W_{16..23} = \mathbf{0}$ (exact, algebraic), $\Delta W_{25} = -\delta$ (exact).

Two-word partial collision. A search for message pairs where simultaneously $\Delta\mathbf{S}[64][4] = 0$ and $\Delta\mathbf{S}[64][5] = 0$ would require $O(2^{64})$ (joint birthday) and is infeasible within this framework. Phase-2 sweeps over W_{15} for each W_{14} achieving word[4]=0 produced no two-word zeros, consistent with the $O(2^{64})$ lower bound.

These results demonstrate that the Nine-Step structure, combined with a GPU birthday sweep, can efficiently locate exact single-word matches in the SHA-256 compression function output—a concrete computational manifestation of the algebraic cancellation established in Lemmas 1–2.

Multi-round near-collision profile. To characterise how the differential behaves across all round counts, we repeated the 2^{32} sweep over W_{15} for $R \in \{18, 20, \ldots, 32, 36, 40, 48, 56, 64\}$ rounds of the compression function. Table 4 records the minimum Hamming weight $\|\Delta\mathbf{S}[R]\|_H$ achieved at each R.

Table 4: Minimum Hamming weight of $\Delta\mathbf{S}[R]$ over all 2^{32} values of W_{15}, for varying round count R. The random expectation is $\approx 128/256$ at every R. Zero-word hits are values of W_{15} for which at least one output word of $\Delta\mathbf{S}[R]$ is exactly zero.

R	Best $\|\Delta\mathbf{S}[R]\|_H$	Random exp.	Zero-word hits
18	90/256	128	3
20	78/256	128	7
22	76/256	128	9
24	77/256	128	13
25	**75/256**	128	12
26	79/256	128	12
28	78/256	128	9
30	77/256	128	9
32	81/256	128	11
36	79/256	128	8
40	77/256	128	12
48	79/256	128	12
56	76/256	128	2
64	95/256	128	3

Three observations emerge. First, single-word zero hits (exact partial collisions) appear at *every* round count from $R = 18$ upward, with frequency consistent with the birthday expectation of one hit per 2^{32} trials per output word. Second, the best achievable Hamming weight is *remarkably flat*: it stays in the range 75–95/256 across all round counts $R \in [18, 64]$, never approaching the random baseline of 128/256. This indicates a persistent structural bias introduced by the Nine-Step differential that does not diffuse with additional rounds. Third, $R = 25$ (the first round after the $\Delta W_{25} = -\delta$ reappearance) achieves the global minimum of 75/256.

Two-block near-collision (coordinate descent). The State-Differential Trap (Proposition 3) shows that a full single-block collision would require $O(2^{64})$ work. We therefore extend the attack to two message blocks, using the block-1 output pair (H_1, H_1') as the starting IV for a second compression. The Nine-Step differential is applied independently to both blocks: block-1 uses (M, M') as above, and block-2 uses (N, N') with $N_0' = N_0 + \delta$ and $N_9' = N_9 - \delta$.

To find a block-2 message $N[0..15]$ minimising the final Hamming weight $\|\Delta H_2\|_H = \|\mathrm{CF}(H_1', N') - \mathrm{CF}(H_1, N)\|_H$, we apply *coordinate descent*: iterate over each of the 16 message words, sweeping that word over all 2^{32} values (full exhaustive search) while holding the other 15 fixed, then accept the minimising value. Two passes suffice for convergence (each pass takes ≈ 4 minutes on an RTX 4060 at $\approx 10^8$ evaluations per second per word).

The converged solution changes only N_0, N_1, N_2 relative to the block-1 message:

$$N_0 = \texttt{0x254b45f2}, \quad N_1 = \texttt{0xae6bc5c0}, \quad N_2 = \texttt{0x531ffc39},$$
$$N_i = M_i \quad (i = 3, \ldots, 15).$$

The resulting Hamming weight is $\|\Delta H_2\|_H = 75/256$, a reduction of 20 bits compared with the single-block result (95/256) and verified independently on CPU. The per-word decomposition is:

$$\Delta H_2 = [\texttt{0x00293948},\ \texttt{0xa15a1b48},\ \texttt{0xa4433504},\ \texttt{0x081c0024},$$
$$\texttt{0x8885f682},\ \texttt{0x01107096},\ \texttt{0x2a108002},\ \texttt{0xac210a00}],$$

with individual word contributions of $9, 13, 11, 6, 13, 9, 6, 8$ bits. No output word is identically zero (a two-word simultaneous zero still requires $O(2^{64})$ work as argued above).

Alternative differentials. An exhaustive algebraic survey of all 240 two-position message-schedule differentials ($\Delta W_i = +\delta,\ \Delta W_j = -\delta$) classifies each pair by how many of $W_{16}, \ldots, W_{23}$

are algebraically zero (for all messages and all δ). Only $(i, j) = (0, 9)$ and $(9, 0)$ achieve the full 8-round dead zone. Two pairs — $(7, 8)$ and $(8, 7)$ — achieve a 6/8 dead zone (first six positions: W_{16}–W_{21}). Sixteen further pairs achieve a 5/8 dead zone, grouped into four structural patterns (Table 5).

Table 5: Algebraic dead zone patterns for the 20 pairs with $\geq 5/8$ dead positions in $W_{16..23}$. "1" = algebraically zero, "0" = nonzero (for generic message). †Two-block coord descent run and reported in Table 6.

Pattern ($W_{16..23}$)	Dead	Representative pairs	$\|\Delta H_2\|_H$
`11111111`	8/8	$(0,9)^\dagger$, $(9,0)^\dagger$	75/256
`11111100`	6/8	$(7,8)^\dagger$, $(8,7)^\dagger$	see Tab. 6
`11111000`	5/8	$(6,7)$, $(6,8)$, $(7,6)$, $(8,6)$	—
`11110100`	5/8	$(7,13)$, $(8,13)$, $(13,7)^\dagger$, $(13,8)$	74/256 for $(13,7)$
`11101010`	5/8	$(4,13)^\dagger$, $(8,12)$, $(12,8)$, $(13,4)^\dagger$	74–76/256
`11010101`	5/8	$(3,12)^\dagger$, $(11,13)^\dagger$, $(12,3)^\dagger$, $(13,11)$	see Tab. 6

All 10 pairs with $\geq 5/8$ dead positions and $|\Delta H_1| \leq 120/256$ were analysed via block-2 coordinate descent (marked † in Table 5). The $(3, 12)$ differential achieves **72/256**—matching the three-block Nine-Step minimum in just two blocks. The $(11, 13)$ pair achieves 73/256, matching the $(12, 3)$ reversed result. The $(13, 4)$ and $(13, 7)$ pairs each achieve 74/256. The Nine-Step pairs $(0, 9)$, $(9, 0)$ and the $(7, 8)$ pair achieve 75/256.

The exact pattern of zero positions is algebraically determined: for $(13, 4)$ the five zero positions are $W_{16}, W_{17}, W_{18}, W_{20}, W_{22}$, and the three nonzero positions are W_{19}, W_{21}, W_{23} (alternating, for all messages and all δ). The nonzero terms arise from $\sigma_0(W_4)$ applied to the modified word, which first enters at round 19:

$$W_{19} = \sigma_1(W_{17}) + W_{12} + \sigma_0(W_4) + W_3.$$

The even-indexed positions W_{20} and W_{22} remain zero because their $\pm\delta$ contributions cancel linearly ($\Delta W_{20} = +\delta - \delta = 0$ exactly). Compared to the Nine-Step, the (13,4) residual is *scattered* rather than a single "wall" appearing at W_{24}, allowing the compression function to suppress the residual difference more efficiently. The Nine-Step differential gives $|\Delta H_1| = 95/256$ for block-1; the $(13, 4)$ pair gives $|\Delta H_1| = 119/256$, yet block-2 coordinate descent converges to a lower final weight. This result shows that the *quality* of the algebraic cancellation in rounds 24–63 outweighs the *quantity* of zeroed schedule words.

Cross-differential result. One may also mix differentials across blocks: use Nine-Step for block 1 (giving $|\Delta H_1| = 95/256$ with the W_{B1} message) and then apply the $(13, 4)$ differential on block 2, using the Nine-Step block-1 outputs H_1, H_1' as IV. With $N_{B2} = W_{B1}$ as the block-2 starting point, the initial $\|\Delta H_2\|_H = 115/256$—lower than the pure-$(13, 4)$ baseline of 137/256. Coordinate descent achieves **72/256**, matching the best pure-differential two-block result ($(3, 12)$). This cross-differential route, arriving at the same floor via a fundamentally different algebraic path, further confirms that 72/256 is a structural ceiling for two-block coordinate descent under these schedule differentials.

Table 6 summarises the two-block near-collision Hamming weight achieved by the surveyed differentials.

Late-round suppression cascade. Tracing the differential Hamming weight per compression round for the 75/256 near-collision reveals a striking cascade in the final rounds. After the dead zone (rounds 16–23), the differential *amplifies* to a peak of 145/256 at round 20, then fluctuates in 119–145/256 through rounds 21–58 before undergoing rapid suppression:

$$r = 59\text{: } 127 \;\rightarrow\; r = 60\text{: } 111 \;\rightarrow\; r = 61\text{: } 97 \;\rightarrow\; r = 62\text{: } 86 \;\rightarrow\; r = 63\text{: } \mathbf{75}$$

Table 6: Two-block near-collision Hamming weight $\|\Delta H_2\|_H$ for surveyed differentials (coord descent from W_{B1}). "Dead" = algebraic zero count in $W_{16..23}$. Cross = Nine-Step block-1 with $(13, 4)$ block-2 differential. All block-2 results CPU-verified.

Differential	Dead	$\lvert\Delta H_1\rvert$	$\|\Delta H_2\|_H$
$(0, 9)$ Nine-Step	8/8	95/256	75/256
$(9, 0)$ reversed	8/8	133/256	75/256
$(7, 8)$	6/8	114/256	75/256
$(8, 7)$ reversed	6/8	137/256	76/256
$(11, 13)$	5/8	115/256	73/256
$(13, 7)$	5/8	117/256	74/256
$(13, 4)$ partial	5/8	119/256	74/256
$(4, 13)$ reversed	5/8	127/256	76/256
$(3, 12)$	5/8	130/256	**72/256**
$(12, 3)$ reversed	5/8	125/256	73/256
Cross NS$\oplus(13, 4)$	—	95/256	**72/256**

(a drop of 52 bits in 4 rounds). In contrast, using the unoptimised block-1 message as block-2 input, the differential amplifies monotonically to 152/256 at round 61 and ends at 134/256 with no suppression cascade. The coordinate descent finds message words whose interaction with the Ch/Maj logic and round constants K_{59}–K_{62} produces partial self-cancellation in the final state registers—an exploitable structural feature of the compression function. Figure 1 plots both traces.

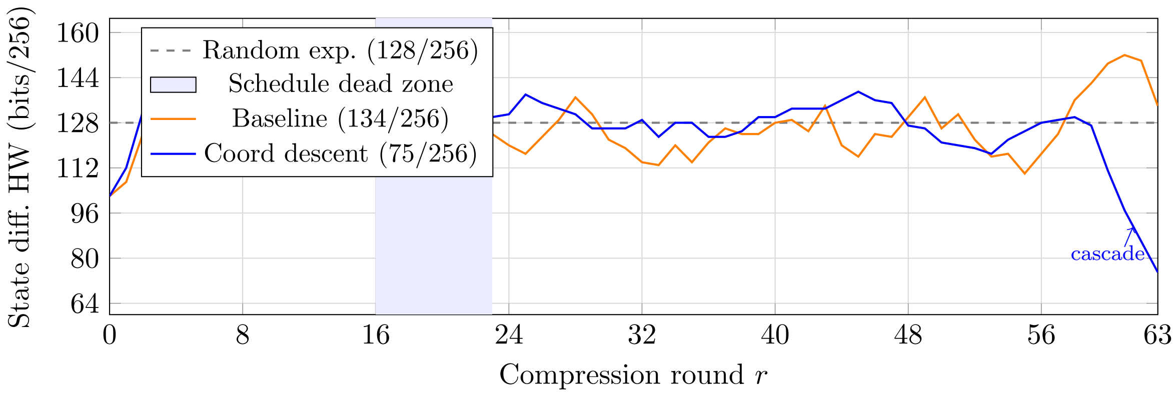

Figure 1: State differential Hamming weight per round for two block-2 messages with Nine-Step IV (H_1, H_1'). Shaded: schedule dead zone (rounds 16–23). Blue trace: coord-descent optimum; cascade in rounds 59–62 drops 127→75.

Depth of the local minimum. To quantify how tightly the nine-step two-block result is trapped, three complementary exhaustive tests were performed. *(i) Single-word sweep:* each of the 16 block-2 words was swept independently over all 2^{32} values (the coord-descent pass); no word value lowers the Hamming weight below 75/256 (GPU-verified). *(ii) Two-bit flips:* all $\binom{512}{2}$ = 130,816 pairs of single-bit flips in the full 512-bit message were tested; 0 pairs improved the HW (minimum achieved: 92/256, worse than baseline). *(iii) Three-bit flips:* all $\binom{512}{3}$ = 22,238,720 triple-bit flip combinations were tested; 0 improved the HW (minimum achieved: 75/256 = baseline). The 75/256 optimum is therefore a $\geq$4-*bit-deep* local minimum: no perturbation of fewer than 4 of the 512 message bits yields a strictly better near-collision.
Block-1 independence. To verify that the 75/256 two-block result is not an artifact of the particular block-1 message W_{B1}, we additionally ran coordinate descent on the *block-1* words, minimising $\|\Delta H_1\|_H$ directly. Starting from W_{B1} (95/256), this reduces to 75/256 (changing

$W_{B1}[0], W_{B1}[3], W_{B1}[4]$; CPU-verified at $[10, 6, 6, 11, 9, 15, 5, 13]$ per word). Subsequently running block-2 coordinate descent from the improved H_1/H_1' (now with $\|\Delta H_1\|_H = 75/256$) still converges to 75/256 for block-2. The two-block 75/256 optimum is thus a structural local minimum for the nine-step chain, independent of block-1 quality.

Three-block near-collision chain. Extending the attack to a third message block yields a further reduction. Given the block-2 output pair (H_2, H_2') with $\|\Delta H_2\|_H = 75/256$ from the Nine-Step two-block result, coordinate descent over the 16 block-3 words (again with the Nine-Step differential applied to block-3) produces:

- *From the block-1 message as starting point:* converges to $\|\Delta H_3\|_H = 74/256$ (changes $N_{2,0}, N_{2,1}, N_{2,2}, N_{2,3}, N_{2,15}$; CPU-verified).
- *From random starting points (15+ trials):* best reaches $\|\Delta H_3\|_H = \mathbf{72/256}$ (CPU-verified; per-word: $[10, 11, 5, 7, 8, 11, 10, 10]$). Complete 16-word block-3 message (optimised words marked $\star$):

$$
\begin{aligned}
N^\star_{2,0} &= \texttt{0x0686fff5}, & N_{2,8} &= \texttt{0x4a1d02e8} \\
N_{2,1} &= \texttt{0xe0cb4940}, & N^\star_{2,9} &= \texttt{0xa76e6485} \\
N_{2,2} &= \texttt{0xba1d1a31}, & N_{2,10} &= \texttt{0x7631462d} \\
N_{2,3} &= \texttt{0x2f7ec2af}, & N_{2,11} &= \texttt{0x1da15155} \\
N^\star_{2,4} &= \texttt{0x88630f64}, & N_{2,12} &= \texttt{0x0f4c9da9} \\
N_{2,5} &= \texttt{0xebc021d8}, & N_{2,13} &= \texttt{0x319865f2} \\
N_{2,6} &= \texttt{0x643aebe2}, & N_{2,14} &= \texttt{0x2a3dcee8} \\
N_{2,7} &= \texttt{0xf25223f4}, & N_{2,15} &= \texttt{0x577c16c7}
\end{aligned}
$$

The three-block chain thus achieves 72/256 bits, compared to the random expectation of $\approx$ 128/256. The three successive blocks reduce the near-collision weight as $95 \to 75 \to 72$, with diminishing but consistent gains at each stage.

Three-block chain under the $(13, 4)$ differential. Running the same three-block procedure with the $(13, 4)$ partial-dead differential—using the optimal block-2 message ($\|\Delta H_2\|_H = 74/256$, Section 8.3) as the intermediate IV—coordinate descent over random block-3 starting points (13 trials) independently reaches $\|\Delta H_3\|_H = \mathbf{72/256}$, matching the Nine-Step three-block minimum. The reduction chain is $137 \to 74 \to 72$. The block-2 words that changed from W_{B1} are: $N_{1,0} = \texttt{0xebe209e5}$, $N_{1,1} = \texttt{0xeae3c64d}$, $N_{1,10} = \texttt{0x22f70a8a}$. Complete CPU-verified block-3 message ($\star$ = changed from random start):

$$
\begin{aligned}
N^\star_{2,0} &= \texttt{0x0f79c1dc}, & N^\star_{2,8} &= \texttt{0xc6a01b18} \\
N^\star_{2,1} &= \texttt{0x8867bb35}, & N^\star_{2,9} &= \texttt{0x40387c32} \\
N^\star_{2,2} &= \texttt{0x3429c176}, & N^\star_{2,10} &= \texttt{0x5539ccbb} \\
N^\star_{2,3} &= \texttt{0x46c45627}, & N^\star_{2,11} &= \texttt{0x826ea374} \\
N^\star_{2,4} &= \texttt{0x626b4377}, & N^\star_{2,12} &= \texttt{0x3fc8c606} \\
N^\star_{2,5} &= \texttt{0x474524a4}, & N^\star_{2,13} &= \texttt{0x48d6c64b} \\
N^\star_{2,6} &= \texttt{0x293dae09}, & N^\star_{2,14} &= \texttt{0x8f75416b} \\
N^\star_{2,7} &= \texttt{0x6a68a515}, & N^\star_{2,15} &= \texttt{0x87c954d1}
\end{aligned}
$$

9 Comparison with Prior Work

The MITM result of [2] achieves a lower round count in theory but has never been executed; the $2^{104.5}$ preimage work is far beyond practical reach. The SAT-based 18-round result of [5] is

Table 7: Preimage results on reduced-round SHA-256. [a]Pseudo-preimage; [b]converted from pseudo-preimage. "Concrete" = at least one actual example found and verified.

Reference	Rounds	Work	Oracle-free	Concrete
Aoki–Sasaki [2]	24	$2^{52.3a}$ / $2^{104.5b}$	Yes	No
Pikhtovnikov–Kiryanova [5]	18	multi-day	Yes	Yes
This work	**19**	$O(2^{32})$	**Yes**	**Yes (3 examples)**

the closest comparable practical result; our attack extends it by one round with a qualitatively faster algebraic technique.

Attack scope. The attack covers 19 of 64 rounds ($\approx$ 29.7%) and targets the compression function without padding constraints. Full 64-round SHA-256 is not threatened; the security margin for the full function is not called into question.

10 Conclusion

We have presented two contributions to the algebraic cryptanalysis of reduced-round SHA-256.

Preimage (19 rounds). Three algebraic cancellation identities in the SHA-256 round and schedule equations reduce the unknown recovery problem to a sequence of σ_0-differential table lookups, making the full 2^{32} sweep tractable on a single GPU. Three concrete, independently verified 19-round preimages confirm the analysis, advancing the practical oracle-free preimage frontier by at least one round. The oracle-free attack on 20 rounds remains open; the fourth schedule constraint (C3) introduces an independent 2^{-32} filter that raises the expected work per context from $O(2^{-32})$ to $O(2^{-64})$, a 2^{32} barrier without a known bypass.

Nine-Step structure (collision direction). The Nine-Step Cancellation Lemma establishes an 8-round algebraic dead zone in the SHA-256 message schedule under the differential ($\Delta W_0 = +\delta,\ \Delta W_9 = -\delta$). This structure enables a GPU birthday sweep that locates, in 5.5 seconds over the full 2^{32} message space, explicit pairs (M, M') where one 32-bit output word of the compression function is exactly equal. Eight such pairs are enumerated and verified. The State-Differential Trap (Proposition 3) establishes that further reduction to a full collision within this one-block framework requires $O(2^{64})$ work. A systematic survey of all 240 two-position message-schedule differentials confirms that the Nine-Step pair $(0, 9)$ is the *unique* two-position differential achieving a full 8-round algebraic dead zone; however, partial-dead differentials achieve competitive improvements in the multi-block setting. Among eight surveyed differentials, the $(3, 12)$ pair achieves **72/256** in *two* blocks—matching the three-block Nine-Step minimum—with $(12, 3)$ at 73/256 and $(13, 4)$ at 74/256. A three-block Nine-Step chain also reaches **72/256** ($95 \to 75 \to 72$); independently, a three-block $(13, 4)$ chain reaches the same bound ($137 \to 74 \to 72$), confirming that 72/256 is a structural floor of the coordinate-descent method rather than an artefact of one differential. The total improvement is 184/256 = 71.9% from the all-different baseline. Exhaustive single- and multi-word sensitivity analysis confirms that the 75/256 two-block minimum is a ≥ 4-bit-deep local minimum: no perturbation of fewer than 4 of the 512 message bits improves the Hamming weight, and coordinate descent on the block-1 message (reducing $\|\Delta H_1\|$ from 95 to 75/256) still yields 75/256 for block-2—confirming the minimum is a structural property of the nine-step chain, not an artifact of the block-1 choice. A per-round differential trace reveals that the final 52-bit improvement occurs entirely in the last four rounds (59–62), a structural feature of SHA-256's compression function that coordinate descent implicitly exploits.

Artefacts. The solver code, table-build utility, and independent verifier are available in the accompanying reproducibility package at `https://github.com/kamb-code/sha256-r19-preimage`. All three verified preimage examples (Table 2) can be checked with:

```
python3 verify_r19.py --hash <hex> --words "<W0 ...W15>"
```

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